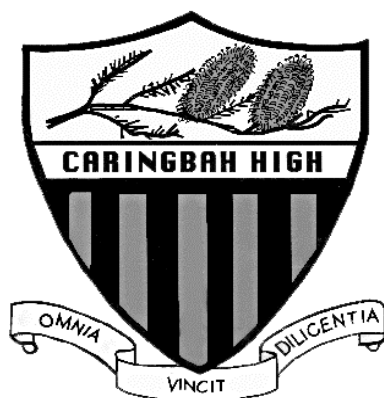




Caringbah High School

Year 11 2018

Mathematics Extension 1

Preliminary Course

Assessment Task 3

General Instructions

- Reading time – 5 minutes
- Working time – 90 minutes
- Write using black pen.
- NESA calculators may be used.
- The approved board of studies reference sheet has been provided.
- In questions 11 - 15, show relevant mathematical reasoning and/or calculations.
- Marks may not be awarded if sufficient reasons and/or working have not been shown.

Total marks – 65

Section I **10 marks**

Attempt Questions 1-5

Mark your answers on the Multiple Choice answer sheet provided. You may detach the sheet and write your name on it.

Section II **55 marks**

Attempt Questions 11-15

Write your answers in the answer booklets provided.

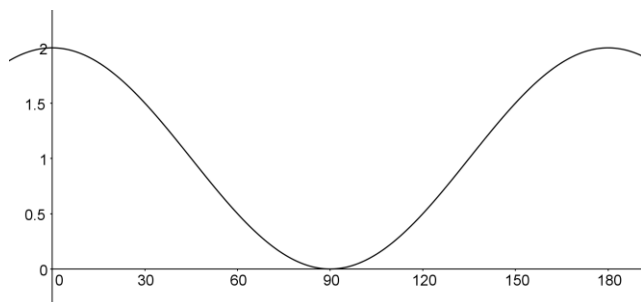
Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only							
Section I	Section II					Total	
Q 1-10	Q11	Q12	Q13	Q14	Q15		
/10	/11	/11	/11	/11	/11	/65	%

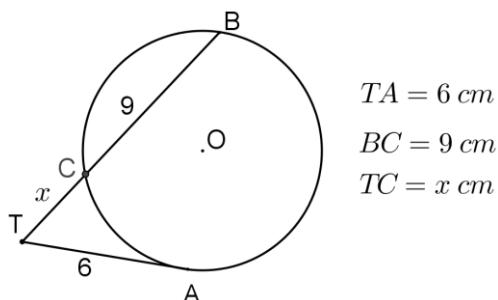
4. A possible equation for this curve is:



- (A) $y = -1 + \cos 2x$ (B) $y = 1 + \cos 2x$
(C) $y = \cos(2x + 1)$ (D) $y = 1 - \cos 2x$
5. What is the acute angle between the lines $y = 7 - 4x$ and $2x - 3y - 6 = 0$, to the nearest degree?
- (A) 26° (B) 48°
(C) 70° (D) 75°
6. An expression for the general solution to the trigonometric equation $\tan 3x = -\sqrt{3}$, where n is any integer and x is in degrees is:

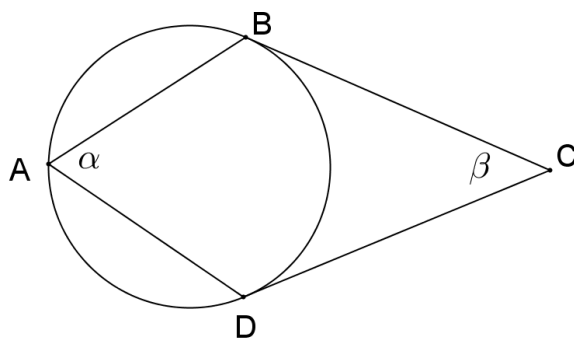
- (A) $x = 60n - 20$ (B) $x = 60n - 60$
(C) $x = 60n + 60$ (D) $x = 60n + 20$

7. The line TA is a tangent to the circle at A . TB is a secant cutting the circle at B and C .



What is the value of x ?

- (A) 2 (B) 3
(C) 4 (D) 12
8. The solution to the inequality $6 - x - x^2 \leq 0$ is:
- (A) $x \leq -3$ or $x \geq 2$ (B) $-3 \leq x \leq 2$
(C) $x \leq -2$ or $x \geq 3$ (D) $-2 \leq x \leq 3$
9. What is the correct condition for $y = mx + b$ to be a tangent to $x^2 = 4ay$?
- (A) $am - b = 0$ (B) $am^2 - b = 0$
(C) $am + b = 0$ (D) $am^2 + b = 0$
10. In the diagram below, BC and DC are tangents.



Which statement below is correct?

- (A) $\alpha + \beta = 180^\circ$ (B) $2\alpha + \beta = 180^\circ$
(C) $\alpha + 2\beta = 180^\circ$ (D) $2\alpha - \beta = 180^\circ$

End of Section I

Section II

60 marks

Attempt Questions 11–15

Allow about 1 hour 15 minutes for this section

Answer each question in a SEPARATE writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (11 marks)

- (a) If α and β are roots of the equation $3x^2 - 6x + 12 = 0$, find the value of
- | | |
|------------------------------------|---|
| i) $\alpha^2\beta + \alpha\beta^2$ | 1 |
| ii) $\alpha^2 + \beta^2$ | 1 |
- (b) For the parabola $2y = x^2 - 4x$,
- | | |
|---|---|
| i) Find the coordinates of the vertex. | 1 |
| ii) Find the coordinates of the focus. | 1 |
| iii) Sketch the curve, clearly showing the vertex, focus and the directrix. | 1 |
- (c) Differentiate $f(x) = x^2 + 3x - 1$ from first principles. 2
- (d) Find $\lim_{n \rightarrow \infty} \frac{3n^2 + 2n - 1}{5n^2 + 4}$ 1
- (e) Find the equation of the normal to the curve $y = \frac{8}{x+3}$ at the point $(1, 2)$. 3

End of Question 11

Question 12 (11 marks) Use a SEPARATE writing booklet.

- (a) Solve the inequality $\frac{2x - 3}{x - 4} \geq -3$ 2
- (b) Shade the region on the number plane where the following inequalities hold simultaneously. (Draw a $\frac{1}{3}$ page diagram.) 2
- $$y \leq x^2 - 4 \quad \text{and} \quad y > 4 - x^2$$
- (c) Find values for b and c such that $x^2 - x \equiv (x + 3)^2 + bx + c$. 2
- (d) Express $\sqrt{3} \cos \theta - \sin \theta$, in the form $R \cos(\theta + \varphi)$, 2
where $R > 0$ and $0^\circ < \varphi < 90^\circ$.
- (e) $f(x) = \begin{cases} |x + 1|, & \text{for } x < 0 \\ a^x + b, & \text{for } x \geq 0 \end{cases}$
- i) Find the values of a and b if $f(0) = 4$ and $f(1) = 7$. 2
- ii) Evaluate $f(-3) + f(2)$ 1

End of Question 12

Question 13 (11 marks) Use a SEPARATE writing booklet.

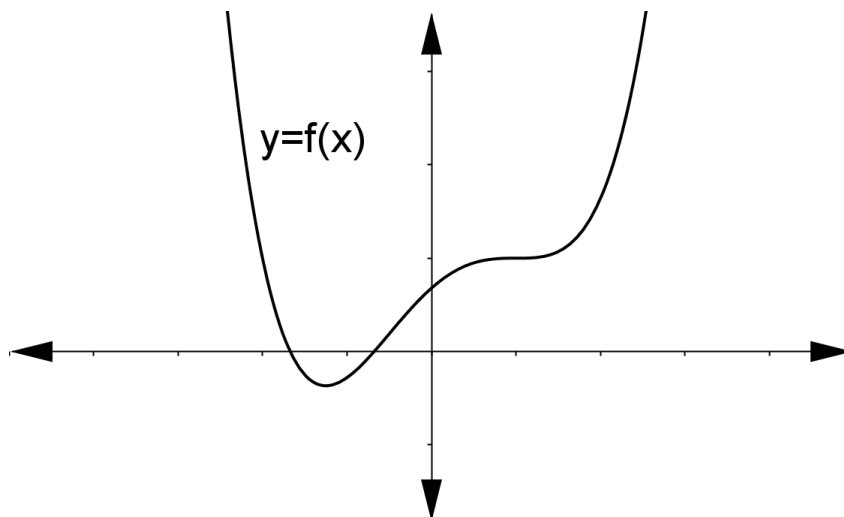
- (a) Let A be the point $(-3, 8)$ and let B be the point $(5, -6)$. Find the coordinates of the point P that divides the interval AB externally in the ratio $3:2$. 1
- (b) Find the equation of the locus of a point, $P(x, y)$, that moves such that its distance from the point $C(2, -3)$ and its distance from the point, $D(-3, 7)$ is in the ratio $2:3$. 2
- (c) Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ 1
- (d) Find the natural domain of the function $f(x) = \sqrt{2x^2 - 9x + 4}$ 2
- (e) Consider the function $f(x) = \frac{9 - x^2}{x^2 - 4}$
- i) Show that the function is even. 1
- ii) State the equations of the asymptotes. 2
- iii) Sketch the graph of $y = f(x)$. (Draw a $\frac{1}{3}$ page diagram) 2

End of Question 13

Question 14 (11 marks) Use a SEPARATE writing booklet.

- (a) The curves $y = 2x^3 + 3$ and $y = x^2 - 2x + 6$ meet at the point $(1, 5)$.
Find the acute angle between these curves at this point of intersection. 3

- (b) The graph of $y = f(x)$ is given below 2



Copy this diagram of $y = f(x)$ onto your answer sheet.
On the same set of axes or directly below it draw a neat sketch of $y = f'(x)$.

- (c) Form a Cartesian equation by eliminating the parameter, θ , from these parametric equations. 2

$$x = 3 \tan \theta \quad y = 2 \sec \theta$$

- (d) Consider the expression $(m - 1)x^2 + (m + 1)x + (m - 1)$

- i) Show that the discriminant in terms of m is 1

$$\Delta = -3m^2 + 10m - 3$$

- ii) Hence find the values of m for which the expression is positive for all values of x . 3

End of Question 14

Question 15 (11 marks) Use a SEPARATE writing booklet.

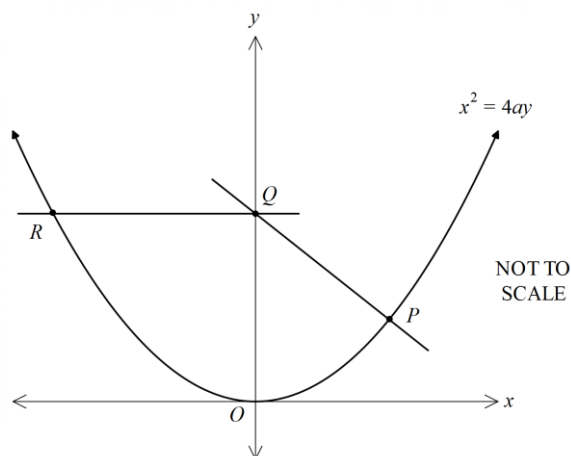
(a) Show that $\sin A \cos A \cos 2A = \frac{1}{4} \sin 4A$ 1

(b) i) Solve the equation $\cos 2\theta = 0.5$, for $0^\circ \leq \theta \leq 360^\circ$ 2

ii) Solve the equation $\sin 2\theta = \sqrt{2} \cos \theta$, for $0^\circ \leq \theta \leq 360^\circ$ 2

(c) Find the equation of the tangent to the curve $y = 2x^3 - 5x + 1$ at the point $A(2, 7)$. 2

- (d) The diagram below shows the graph of the parabola $x^2 = 4ay$.
The normal to the parabola at the variable point $P(2ap, ap^2)$, $p > 0$, cuts the y -axis at Q .
Point R lies on the parabola.



i) Find the coordinates of R given that QR is parallel to the x -axis and $\angle PQR > 90^\circ$. 1

ii) Let M be the midpoint of RQ . Find the Cartesian equation of the locus of M . 3

End of Question 15

End of Examination

Multiple Choice Answer Sheet

Name _____

Colour the oval A, B, C or D that best answers the question.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A B C D

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A  B  C  D 

correct

- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

2018 MATHEMATICS EXT 1 AT3

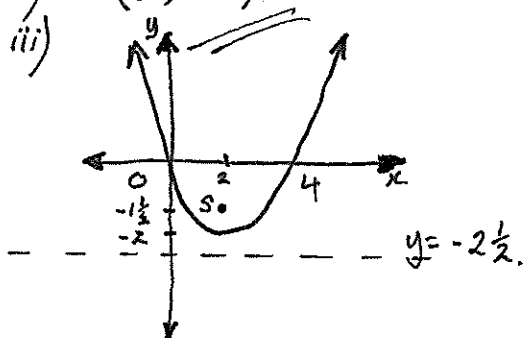
- 1) A 2) D 3) B 4) B 5) C
6) A 7) B 8) A 9) D 10) B

Question 11

a) i) $\alpha + \beta = 2$ $\alpha\beta(\alpha + \beta) = 8$
 $\alpha\beta = 4$

ii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= -4$

b) $2y = (x^2 - 4x + 4) - 4$
 $2(y+2) = (x-2)^2$ $4a = 2$
i) $V(2, -2)$ $a = \frac{1}{2}$
ii) $S(2, -1\frac{1}{2})$



c) $f(x) = x^2 + 3x - 1$
 $f(x+h) = (x+h)^2 + 3(x+h) - 1$
 $= x^2 + 2xh + h^2 + 3x + 3h - 1$
 $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 + 3x + 3h - 1) - (x^2 + 3x - 1)}{h}$
 $= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h}$
 $= \lim_{h \rightarrow 0} (2x + h + 3)$
 $= 2x + 3$

d) $\lim_{n \rightarrow \infty} \frac{\frac{3n^2}{n^2} + \frac{2n}{n^2} - \frac{1}{n^2}}{\frac{5n^2}{n^2} + \frac{4}{n^2}}$
 $= \frac{3}{5}$

e) $y = 8(x+3)^{-1}$
 $y' = -8(x+3)^{-2}$
 $x=1, m_T = \frac{-8}{16}$

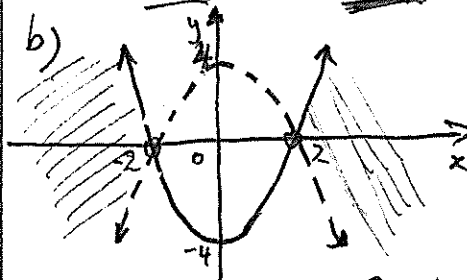
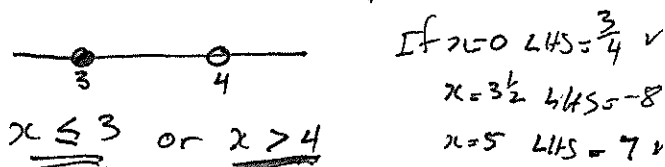
$= -\frac{1}{2} \therefore M_N = +2$

Eq'n of normal
 $(y - y_1) = m(x - x_1)$
 $(y - 2) = 2(x - 1)$
 $y = 2x$

Question 12

a) $\frac{2x-3}{x-4} \geq -3$ $x \neq 4$

critical pts $2x-3 = -3x+12$
 $5x = 15$
 $x = 3$



c) $x^2 - x \equiv (x+3)^2 + bx + c$
If $x=0$
 $0 \equiv 9 + c \therefore c = -9$
If $x=-3$
 $12 \equiv -3b - 9 \therefore b = -7$

d) $\sqrt{3} \cos \theta - \sin \theta$
 $R \cos \theta \cos \phi - R \sin \theta \sin \phi$

$R \cos \phi = \sqrt{3}$ $R \sin \phi = 1$
 $1^2 + (\sqrt{3})^2 = R^2 \therefore R = 2$
 $\tan \phi = \frac{1}{\sqrt{3}} \therefore \phi = 30^\circ$



$2 \cos(\theta + 30^\circ)$

e) i) $x=0$ $a^0 + b = 4 \therefore b=3$
 $x=1$ $a^1 + b = 7 \therefore a=4$
 ii) $f(3) + f(2) = 2 + (4^2 + 3)$
 $= 21$

Question 13

a) $A(-3, 8)$ $B(5, -6)$ $m_{AB} = 3 : -2$
 $P\left(\frac{3 \times 5 + 2 \times 3}{3-2}, \frac{3 \times -6 + 2 \times 8}{3-2}\right)$
 $P(21, -34)$

b) $\frac{d_1}{d_2} = \frac{2}{3}$ so $9d_1^2 = 4d_2^2$

$$9[(x-2)^2 + (y+3)^2] = 4[(x+3)^2 + (y-7)^2]$$

$$9[x^2 - 4x + 4 + y^2 + 6y + 9] = 4[x^2 + 6x + 9 + y^2 - 14y + 49]$$

$$9x^2 - 36x + 36 + 9y^2 + 54y + 81 = 4x^2 + 24x + 36 + 4y^2 - 56y + 196$$

$$5x^2 - 60x + 5y^2 + 110y = 115$$

$$x^2 - 12x + y^2 + 22y = 23$$

$$(x^2 - 12x + 36) + (y^2 + 22y + 121) = 23 + 36 + 121$$

$$(x-6)^2 + (y+11)^2 = 180$$

c) $\lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)} = 4$

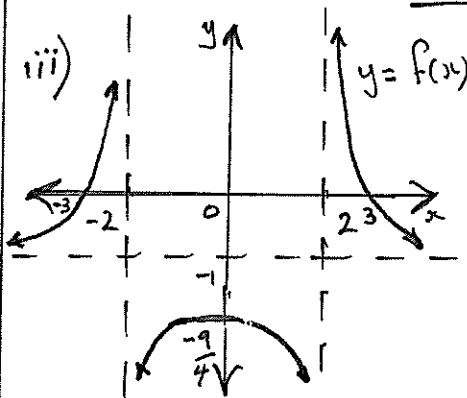
d) $2x^2 - 9x + 4 \geq 0$
 $(2x-1)(x-4) \geq 0$

$x \leq \frac{1}{2}$ or $x \geq 4$

e) i) $f(a) = \frac{9-a^2}{a^2-4}$ $f(-a) = \frac{9-(-a)^2}{(-a)^2-4}$
 $= \frac{9-a^2}{a^2-4}$
 $= f(a)$
 $\therefore f(x)$ is even.

ii) $x^2 - 4 \neq 0 \therefore x \neq 2, x \neq -2$
 as $x \rightarrow \infty$ $\frac{9x^2-1}{1-\frac{4}{x^2}} \rightarrow -1$

$\therefore$ asymptotes $x = \pm 2, y = -1$



Question 14

a) $y = 2x^3 + 3$

$$y' = 6x^2$$

at $x=1$ $m_1 = 6$

$y = x^2 - 2x + 6$

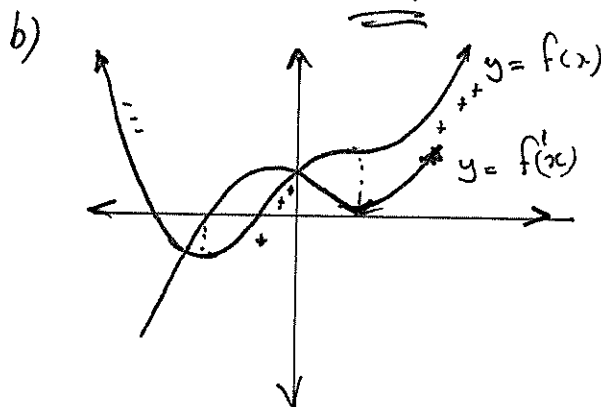
$$y' = 2x - 2$$

$m_2 = 0$

$$\tan \theta = \left| \frac{6-0}{1+6 \times 0} \right|$$

$$= 6$$

$$\theta = 80^\circ 32'$$



c) $\tan \theta = \frac{x}{3}$
 $\sec \theta = \frac{y}{2}$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \frac{x^2}{9} = \frac{y^2}{4}$$

$$\frac{x^2}{9} - \frac{y^2}{4} = -1$$

$$d) i) (m-1)x^2 + (m+1)x + (m-1)$$

$$\Delta = b^2 - 4ac$$

$$= (m+1)^2 - 4(m-1)(m-1)$$

$$= m^2 + 2m + 1 - 4m^2 + 8m - 4$$

$$= -3m^2 + 10m - 3$$

$$ii) \text{ need } a > 0 \quad \Delta < 0$$

$$m-1 > 0 \quad -3m^2 + 10m - 3 < 0$$

$$\underline{m > 1} \quad 3m^2 - 10m + 3 > 0$$

$$(3m-1)(m-3) > 0$$

$$\begin{array}{ccc} \leftarrow 0 & & 0 \rightarrow \\ \frac{1}{3} & & 3 \end{array}$$

$$m < \frac{1}{3} \text{ or } m > 3$$

$$\text{but as } m > 1$$

$$\text{solution is } \underline{m > 3}$$

Question 15

$$a) \sin 2A = 2 \sin A \cos A$$

$$\text{so } \sin A \cos A \cos 2A = \frac{1}{2} \sin 2A \cos 2A$$

$$\sin 4A = 2 \sin 2A \cos 2A$$

$$\text{so } \frac{1}{2} \sin 2A \cos 2A = \frac{1}{2} \times \frac{1}{2} \sin 4A$$

$$= \frac{1}{4} \sin 4A$$

$$b) ii) \sin 2\theta = \sqrt{2} \cos \theta \quad 0^\circ \leq \theta \leq 360^\circ$$

$$2 \sin \theta \cos \theta - \sqrt{2} \cos \theta = 0$$

$$\sqrt{2} \cos \theta (\sqrt{2} \sin \theta - 1) = 0$$

$$\cos \theta = 0 \quad \text{or} \quad \sin \theta = \frac{1}{\sqrt{2}}$$

$$\therefore \theta = 90^\circ, 270^\circ, 45^\circ, 135^\circ$$

$$c) y = 2x^3 - 5x + 1$$

$$y' = 6x^2 - 5 \quad \text{at } x=2 \quad m=19$$

$$\text{Eqn } (y-7) = 19(x-2)$$

$$\underline{y = 19x - 31}$$

d) From ret. sheet

$$i) \text{ Eq'n of normal: } x + py = ap^3 + 2ap$$

$$x=0 \quad y = \frac{ap^3 + 2a}{1}$$

$$\therefore x^2 = 4a^2 p^2 + 8a^2$$

$$= 4a^2(p^2 + 2)$$

$$x = \pm 2a \sqrt{p^2 + 2}$$

$$\therefore R(-2a \sqrt{p^2 + 2}, ap^2 + 2a)$$

$$ii) M(-a \sqrt{p^2 + 2}, ap^2 + 2a)$$

$$x = -a \sqrt{p^2 + 2} \quad y = a(p^2 + 2)$$

$$p^2 + 2 = \frac{y}{a}$$

$$x = -a \sqrt{\frac{y}{a}}$$

$$x^2 = \frac{a^2 y}{a}$$

$$\underline{x^2 = ay} \quad (\text{for } x < 0)$$

$$b) \cos 2\theta = 0.5 \quad 0^\circ \leq \theta \leq 360^\circ$$

$$2\theta = 60, 300, 420, 660 \quad 0^\circ \leq 2\theta \leq 720^\circ$$

$$\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$